

Research on Music Feature Extraction and Machine Learning Classification Algorithm for Anhui Folk Songs

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Abstract—Anhui folk songs are an important part of Chinese traditional music, with unique melodic, rhythmic and tonal characteristics, carrying rich regional culture and historical connotations. In recent years, with the advancement of cultural digitization process, the systematic classification and analysis of Anhui folk songs using modern digital technology has become an important topic in folk song research. In this paper, for the audio signal characteristics of Anhui folk songs, a set of extraction methods based on time domain, frequency domain and high-level music features are proposed, and combined with a variety of machine learning classification algorithms (e.g., support vector machine, decision tree, neural network, etc.) to carry out classification research. By listing the classification effects of different algorithms, the effectiveness and advantages of machine learning technology in the classification of Anhui folk songs are demonstrated, and feasible paths to optimize the classification accuracy are proposed. The research results not only open up new ideas for the digital protection and dissemination of Anhui folk songs, but also provide a reference for the automated classification research of other ethnic music.

Keywords—Anhui folk songs, music feature extraction, time domain features, frequency domain features, machine learning, classification algorithms, support vector machine, neural networks

I. INTRODUCTION

Anhui, which spans the three major water systems of the Yangtze River, the Huai River and the Xin'an River, has a favorable geographical location and is one of the important birthplaces of Chinese prehistoric culture, with outstanding people and rich cultural heritage. [1] Because it is located in the middle of the Yellow River and Yangtze River cultures, it is transitional from language, custom to climate, therefore, Anhui folk songs are characterized by the compatibility of northern and southern styles, [2] with both the roughness and boldness of the north and the gentleness and softness of the south. [3] Anhui folk songs are diverse and colorful, including mountain songs, bugles, ditties, flower drums and lantern songs. [2] According to the geographical distribution, Anhui folk songs can be divided into five color areas: 1. Northern Anhui folk songs: rough and simple, bright and fast, such as: "Pomegranate Picking", "Big Drum Cymbal". 2. Southern Anhui folk songs: delicate and sweet, pure and elegant. 3. For example: "Cattle Flower Drums", "A Small Umbrella Bright Clank". 3. Dabie Mountain Folk Songs: Euphemism and lyricism in minor key, mountain songs are loud and clear, such as: "Picking Tea", "Lang in the mountains singing mountain songs". 4. Chaohu Lake Folk Songs: bold and lyrical, such as:

"Aunt and Sister-in-law to the flowers", "Singing the Four Seasons". 5. Fengyang Folk Songs: passionate and exuberant, dance while singing, such as: "Fengyang Flower Drums", "Fengyang Song". 6. Fengyang Songs: "Fengyang Folk Songs". [4] Anhui folk songs selected for the National Intangible Cultural Heritage Program include: Chaohu Folk Song, Dangtu Folk Song, Dabie Mountain Folk Song, Huizhou Folk Song, Wuhe Folk Song and Fengyang Folk Song. [5]

Anhui folk songs are not only unique in terms of melody, rhythm, key and other musical elements, but also contain deep regional cultural heritage in terms of expression methods and singing styles. However, the traditional music classification method mainly relies on manual experience, which has the problems of low classification efficiency and large error, and is difficult to cope with the needs of organizing and analyzing large-scale music resources.

With the rapid development of artificial intelligence and big data technology, the application of machine learning in the field of music classification and feature extraction has made significant progress. Through machine learning algorithms, key features can be effectively extracted from music signals to achieve automatic classification and analysis of music. Especially in the field of folk music, the use of machine learning technology not only improves the accuracy of classification, but also provides strong technical support for the digitization, protection, inheritance and dissemination of folk music.

This paper focuses on the problem of musical feature extraction and classification of Anhui folk songs, proposes a set of extraction methods based on time domain, frequency domain and high-level musical features, and applies these features to a variety of machine learning algorithms for classification research, not only aiming to improve the accuracy of classification of Anhui folk songs, but also hoping to provide a reference for the research of automatic classification of other ethnic music.

II. MUSICAL CHARACTERISTICS OF ANHUI FOLK SONGS

Due to the different regions, singing times and production and life styles, Anhui folk songs are composed of different and colorful tunes with different feelings. In terms of scales, in addition to the common five-tone and seven-tone scales, there are also six-tone scales and folk songs composed of four-tone and three-tone columns. In terms of tuning, there are five types of tuning: Gong, Shang, Jiao, Zheng, Fei, of which the Zheng

tuning folk song occupies a dominant position in Anhui folk songs, followed by the Fei tuning folk song.[6] Due to the different internal structures and different forms of alternation between tunes, folk songs in different places have their own tuning characteristics. For example, in the western Anhui Dabie Mountains folk songs, in addition to emphasizing the main tone la in the melodic progression, special emphasis is also placed on the genitive tone mi, so that the musical color of the singing is more high-pitched and brighter, while in the Jianghuai region folk songs of the feather-tone type, more emphasis is placed on the subordinate tone re, so that the musical color of the singing is more soft and beautiful.

The rhythms of Anhui folk songs have distinctive folk characteristics, and are generally free and flexible, but in some types of folk songs, they also reflect more regular rhythms. For example, in the type of labor call folk songs, the rhythm is regular, strong and powerful, reflecting the rhythm of collective labor; while in the types of folk songs such as mountain songs and rice-planting songs, the rhythm is more free, changing with the ups and downs of the mood, and there are even some mountain songs that are composed entirely of scattered beats, with no clear laws of strong and weak beats. As shown in Figure 1:

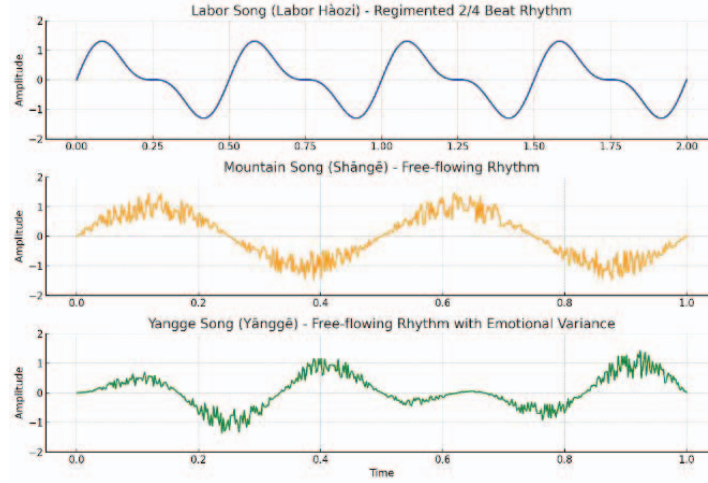


Figure 1 Rhythmic Characteristics of Labor Horns, Mountain Songs, and Yangge of Anhui Folk Songs Waveform Example Chart

In addition, the melodies of Anhui folk songs are often linked to local life and labor scenes, and the melodic rhythms and lyrics are colloquial, with strong expressive and narrative qualities.

III. MUSIC FEATURE EXTRACTION METHOD

Musical feature extraction is a key step to transform the core information in audio signals into feature data that can be used for classification and analysis. For the audio signals of Anhui folk songs, this paper starts from the basic time and frequency domain features, and further deepens the extraction of advanced musical features to better reflect their complex structures such as melody and rhythm. This process includes preprocessing of the audio signal, extraction of basic features and analysis of advanced features, which ultimately provides high-quality input data for the machine learning classification algorithm.

A. Pre-processing of audio signals

In audio signal processing, denoising is the first step to improve the quality of feature extraction. Due to the diversity of recording environments, the audio signals of Anhui folk songs may contain background noise, such as wind, environmental sound or electrical equipment noise. In order to ensure the accuracy of the subsequent feature extraction, the audio is usually denoised using methods such as band-pass filtering, mean filtering or wavelet transform. After removing the external noise, the clarity and accuracy of the audio signal will be greatly improved, which helps to extract more representative music features in the subsequent steps.

The following figure shows the spectrum before and after denoising the 0-5th second music clip in the version of Anhui folk song "Pomegranate Picking" sung by Cao Xinyun and Ma Liuzhu in QQ music, as shown in Figure 2:

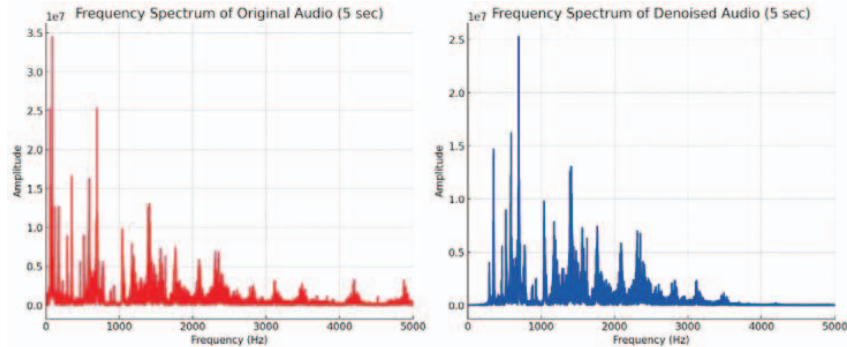


Figure 2 Spectrogram of Anhui folk song "Pomegranate Picking" before and after denoising 5 seconds clip

As can be seen, the left figure (original audio spectrum), showing the frequency distribution of the original audio signal, contains more high-frequency and low-frequency noise, and the right figure (de-noised frequency spectrum) after band-pass filtering and denoising, the high-frequency noise has been reduced, highlighting the main frequency components in the audio, which makes the sound quality clearer.

In order to facilitate the extraction and analysis of the musical features of Anhui folk songs, the audio needs to be clipped. Through shearing, longer audio segments can be decomposed into shorter audio frames to fit the input requirements of machine learning models. Commonly used clipping methods include fixed-length clipping, which divides the audio into equal-length time segments, and dynamic clipping based on musical structure, which is based on the rhythm and melody changes of the music and can better preserve the structural information of the music.

The sampling rate of the audio signal determines the accuracy of signal processing and the efficiency of computation. In the classification task of Anhui folk songs, the sampling rate of the audio usually needs to be adjusted in order to balance the processing speed and the accuracy of feature extraction. A higher sampling rate captures more

details but increases the computational overhead; a lower sampling rate is more computationally efficient but may lose some audio details. In the case of singing, singers will arbitrarily change the melody according to different situations or moods, a basic tune will derive many variants, and singing can be done in either real or falsetto voice without too many limitations, so experiments are needed to determine the optimal parameters to ensure the effectiveness of feature extraction.

In the audio signal processing process, there are obvious differences in the amplitudes of different audio clips. In order to avoid the influence of these differences on feature extraction and classification, the signal needs to be normalized to adjust the amplitude of the audio signal to the same range. Common normalization methods include max-min normalization and Z-score normalization, which can effectively reduce the impact of volume differences on the classification model, so that the model focuses more on the actual musical features of the audio. The following is an example of the 9-19 seconds music clip from the version of the Anhui folk song "Pomegranate Picking" sung by Cao Xinyun and Ma Liuzhu in QQ music, as shown in Figure 3:

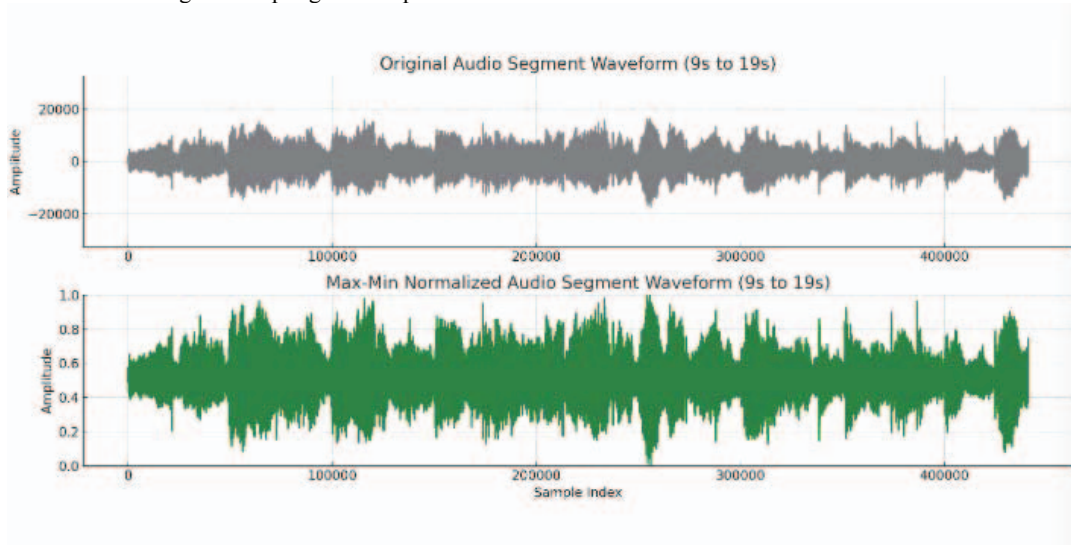


Figure 3 Max-Min normalized waveform of the audio clip from the 9th to the 19th second of the Anhui folk song "Pomegranate Picking"

As can be seen, after Max-Min normalization, the amplitude is compressed to a range of 0 to 1, preserving the relative variation of the waveform, but the amplitude is normalized in order to reduce the effect of volume differences, making it suitable for classification models and feature extraction tasks.

B. Time and frequency domain feature extraction

Short-time energy is one of the most commonly used metrics in time-domain characterization, and is used to describe the energy change of an audio signal within each short time window. Anhui folk songs have more obvious changes in tone intensity in different singing forms, especially in types such as labor horns, where the strong and weak rhythms are obvious. Therefore, short-time energy can effectively capture these dynamic changes. By splitting the audio signal into frames and calculating the energy of each

frame, the intensity changes of the music can be reflected, which plays an important role in distinguishing different types of folk songs.

Short-term energy [7]:

$$E_n = \sum_{m=0}^{N-1} x_n^2(m) \quad (1)$$

The over-zero rate refers to the number of times the audio signal changes from positive to negative or from negative to positive in a time window [8]. This index can reflect the change speed of the audio signal and is suitable for analyzing the rhythmic characteristics of Anhui folk songs. Audio signals with a high rate of zero crossing are usually characterized by fast rhythm or high frequency of sound change, while audio signals with a low rate of zero crossing

indicate smooth melodic lines. Since the rhythms of Anhui folk songs are flexible and changeable, the over-zero rate helps to recognize rhythm-intensive and slow folk songs.

Over-zero rate:

$$ZCR = \frac{1}{N} \sum_{n=1}^{N-1} \delta(x(n) \cdot x(n-1)) \quad (2)$$

In speech recognition technology, the Mel Frequency Cepstrum Coefficient (MFCC) is one of the most common feature extraction methods for speech signals [9]. The MFCC is one of the most widely used frequency domain features in audio classification, and it can effectively describe the spectral envelope of audio signals. By performing Fourier transform on the audio signal, converting the time-domain signal into a frequency-domain signal, and then extracting the spectral cepstrum coefficients on the Meier scale, MFCC can capture the pitch and timbre information perceived by the human ear. In the classification of Anhui folk songs, MFCC can well describe the melody, pitch and other musical features, which is an important basis for distinguishing different folk songs.

Mel-Frequency Cepstral Coefficients (MFCC):

$$MFCC(n) = \sum_{k=1}^K n \log|X(k)| \cdot \cos\left(\frac{\pi n(k-0.5)}{K}\right) \quad (3)$$

The spectral center of mass describes the "center of gravity" of the spectrum of an audio signal, i.e., the center of

the frequency distribution, and is often used to reflect the brightness of a sound. The timbre of different songs in Anhui Folk Songs varies greatly, with some being brighter and others being duller. These timbre differences can be identified by calculating the spectral center of mass. A signal with a higher spectral center of mass usually indicates a brighter sound, while a signal with a lower spectral center of mass indicates a lower, heavier sound. Changes in spectral center of mass can effectively distinguish different styles of Anhui folk songs.

Spectral Centroid:

$$SC = \frac{\sum_f f \cdot |X(f)|}{\sum_f |X(f)|} \quad (4)$$

C. Advanced music feature extraction

Pitch is one of the most basic elements in music, which is directly related to the composition of melody. For pitch extraction in Anhui folk songs, a combination of autocorrelation method and frequency domain analysis is usually used to accurately identify the pitch and variation of notes. By analyzing the period characteristics of the audio signal, the frequency information of pitch can be obtained, which is crucial for distinguishing the melodic characteristics of different Anhui folk songs. Pitch extraction not only helps to understand the melodic direction of Anhui folk songs, but also provides a basis for subsequent harmonic analysis, as shown in Fig. 4:

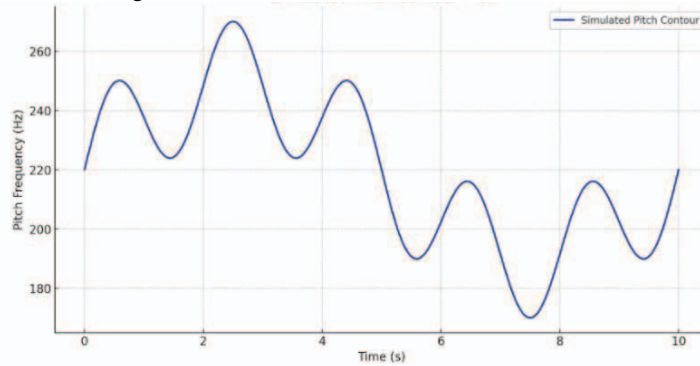


Figure 4 Simulated Pitch Contour Plot

Timbre is an important but often neglected feature in music, which determines the different feelings produced by different instruments or vocals at the same pitch. In order to extract the timbre features of Anhui folk songs, the spectral center of mass, spectral bandwidth and its high-frequency spectral features of the audio signal can be used for comprehensive analysis. The extraction of timbre features helps to identify the differences in sound quality brought by different singers or instruments, thus providing more dimensions for the classification of Anhui folk songs.

Rhythmic pattern refers to the combination of beats and rhythms in music, reflecting the time structure of music. The rhythms of Anhui folk songs are often varied, showing the uniqueness of the local culture. By analyzing the rhythm of audio signals, information such as note duration, interval and accent position can be extracted. Commonly used methods include rhythmic periodicity analysis and beat detection,

which are capable of identifying repetitive rhythms and unique rhythmic patterns in Anhui folk songs, thus providing a basis for the classification and comparison of different types of Anhui folk songs.

IV. CLASSIFICATION ALGORITHM OF ANHUI FOLK SONGS BASED ON MACHINE LEARNING

Support Vector Machine (SVM) is a commonly used supervised learning algorithm, especially for classification problems. It separates sample points of different classes by constructing an optimal hyperplane. In the classification of Anhui folk songs, SVM can effectively handle high-dimensional feature data, especially when the number of samples is small, and the classification performance can be improved by choosing an appropriate kernel function. Using SVM to classify the extracted audio features can achieve a high accuracy rate, which is of great practical value for automatically identifying, organizing and disseminating Anhui folk songs. As shown in Fig. 5:

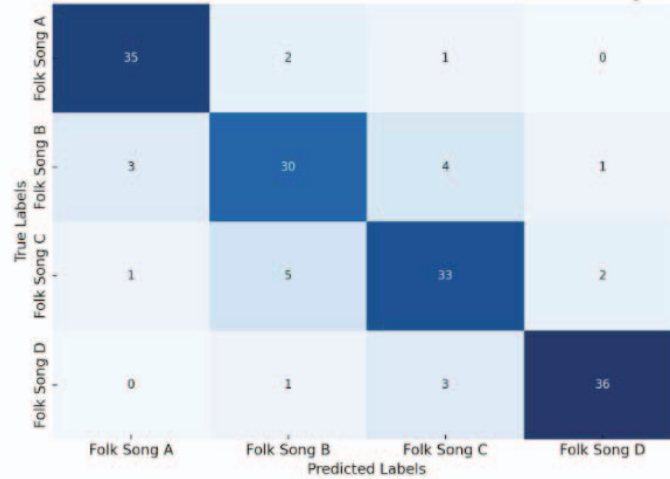


Figure 5 Confusion Matrix for SVM Classification of Anhui Folk Song

This is a simulated confusion matrix chart for Support Vector Machine (SVM) in the classification of Anhui folk songs. This chart shows the classification of different categories of Anhui folk songs, where the values on the diagonal line represent the number of correctly classified samples, and the values on the off-diagonal line indicate the misclassification. This chart allows us to visualize the classification performance of SVM on each category, so that we can further analyze the accuracy and classification effect of the model.

Decision tree is a simple and intuitive classification algorithm that categorizes data by dividing it into different decision paths to form a tree structure. Its strong interpretability and simple implementation make it widely used in music classification. In the classification study of Anhui folk songs, the decision tree can make layer-by-layer decisions based on audio features to accurately distinguish different types of folk songs. By pruning the decision tree, the

overfitting is effectively reduced and the generalization ability of the classification model is improved.

Gini coefficient formula for decision trees [10].

$$gini(T) = 1 - \sum p_i^2 \quad (5)$$

Random forest is an integrated learning method consisting of multiple decision trees that improves classification accuracy and robustness. Overfitting can be prevented by randomly selecting features and samples for training. In the classification of Anhui folk songs, Random Forest takes advantage of its diversity and integration to capture the features of audio signals more comprehensively, and can achieve high classification accuracy in a variety of classification tasks, providing an effective tool for the study of Anhui folk songs, as shown in Fig. 6:

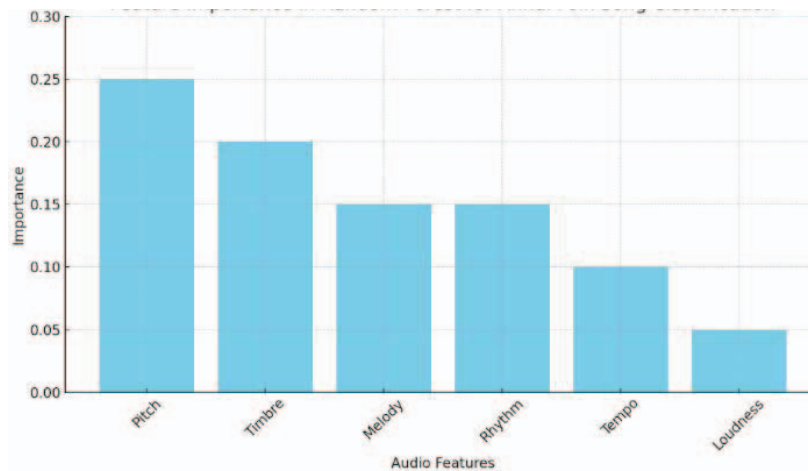


Figure 6 Feature Importance in Random Forest for Anhui Folk Song Classification

In recent years, deep learning techniques have also demonstrated strong capabilities in the field of audio classification, especially in dealing with complex features and large-scale datasets. Convolutional neural network (CNN), as a typical deep learning model, can automatically extract advanced audio features and realize audio classification. In the

classification study of Anhui folk songs, the CNN model can effectively handle multi-dimensional feature data and recognize the nuances of folk songs. Through the training of a large amount of data, this deep learning model has excellent performance in terms of classification accuracy and stability, and injects new vitality into the study of traditional folk songs.

V. CONCLUSION

In this paper, the musical feature extraction and machine learning classification algorithms of Anhui folk songs are studied in depth. Firstly, the musical features of Anhui folk songs, such as melodies, rhythms, modes, etc., are systematically analyzed, and the influence of its unique regional culture and historical background on the musical features is pointed out. On this basis, a set of extraction methods based on time-domain, frequency-domain and high-level musical features is proposed, and combined with a variety of machine learning classification algorithms (e.g., support vector machine, decision tree, neural network, etc.) to carry out classification research. By listing the classification effects of different algorithms, the effectiveness and advantages of machine learning technology in the classification of Anhui folk songs are demonstrated, and feasible paths to optimize the classification accuracy are proposed. The research results not only open up new ideas for the digital protection and dissemination of Anhui folk songs, but also provide a reference for the automated classification research of other ethnic music.

In future research, the musical feature extraction and classification algorithms of Anhui folk songs can be further optimized by combining deep learning techniques. With the development of audio processing technology, future research can also combine advanced feature extraction means such as sentiment analysis to explore the classification method of Anhui folk songs from the perspective of musical emotion. With the help of natural language processing technology and multimodal data fusion, textual information such as lyrics and singing styles can be combined with audio data to construct a more comprehensive classification model.

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